Solution

The expression for z is $z = \overline{AB\overline{C}}$. Use DeMorgan's theorem to break the large inversion sign:

$$z = \overline{A} + \overline{B} + \overline{\overline{C}}$$

Cancel the double inversions over C to obtain

$$z = \overline{A} + \overline{B} + C$$

REVIEW QUESTIONS

1. Use DeMorgan's theorems to convert the expression $z = \overline{(A + B) \cdot \overline{C}}$ to one that has only single-variable inversions.
2. Repeat question 1 for the expression $y = \overline{R\overline{S}T + \overline{Q}}$.
3. Implement a circuit having output expression $z = \overline{A}\,\overline{B}C$ using only a NOR gate and an INVERTER.
4. Use DeMorgan's theorems to convert $y = \overline{\overline{A} + \overline{B} + \overline{C}D}$ to an expression containing only single-variable inversions.

3-12 UNIVERSALITY OF NAND GATES AND NOR GATES

All Boolean expressions consist of various combinations of the basic operations of OR, AND, and INVERT. Therefore, any expression can be implemented using combinations of OR gates, AND gates, and INVERTERs. It is possible, however, to implement any logic expression using *only* NAND gates and no other type of gate. This is because NAND gates, in the proper combination, can be used to perform each of the Boolean operations OR, AND, and INVERT. This is demonstrated in Figure 3-29.

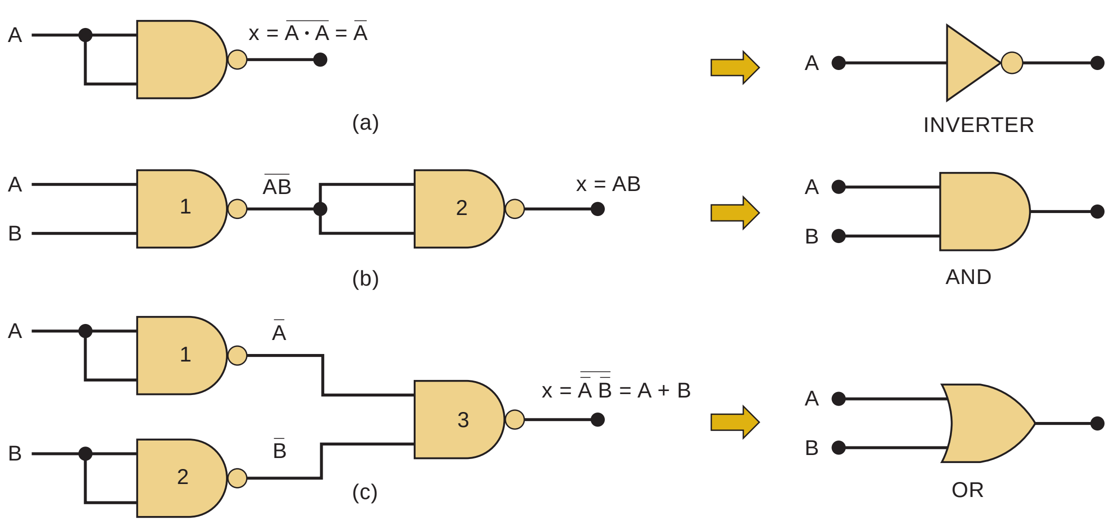

FIGURE 3-29 NAND gates can be used to implement any Boolean function.

First, in Figure 3-29(a), we have a two-input NAND gate whose inputs are purposely connected together so that the variable A is applied to both. In this configuration, the NAND simply acts as INVERTER because its output is $x = \overline{A \cdot A} = \overline{A}$.

In Figure 3-29(b), we have two NAND gates connected so that the AND operation is performed. NAND gate 2 is used as an INVERTER to change $\overline{AB}$ to $\overline{\overline{AB}} = AB$, which is the desired AND function.

The OR operation can be implemented using NAND gates connected as shown in Figure 3-29(c). Here NAND gates 1 and 2 are used as INVERTERs to invert the inputs, so that the final output is $x = \overline{\overline{A} \cdot \overline{B}}$, which can be simplified to $x = A + B$ using DeMorgan's theorem.

In a similar manner, it can be shown that NOR gates can be arranged to implement any of the Boolean operations. This is illustrated in Figure 3-30. Part (a) shows that a NOR gate with its inputs connected together behaves as an INVERTER because the output is $x = \overline{A + A} = \overline{A}$.

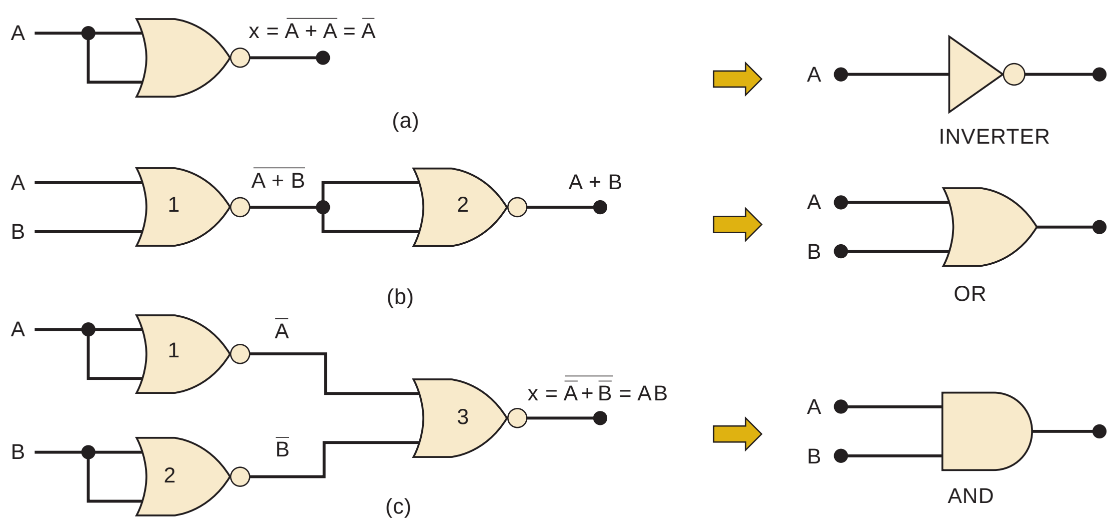

FIGURE 3-30 NOR gates can be used to implement any Boolean operation.

In Figure 3-30(b), two NOR gates are arranged so that the OR operation is performed. NOR gate 2 is used as an INVERTER to change $\overline{A + B}$ to $\overline{\overline{A + B}} = A + B$, which is the desired OR function.

The AND operation can be implemented with NOR gates as shown in Figure 3-30(c). Here, NOR gates 1 and 2 are used as INVERTERs to invert the inputs, so that the final output is $x = \overline{\overline{A} + \overline{B}}$, which can be simplified to $x = A \cdot B$ by use of DeMorgan's theorem.

Since any of the Boolean operations can be implemented using only NAND gates, any logic circuit can be constructed using only NAND gates. The same is true for NOR gates. This characteristic of NAND and NOR gates can be very useful in logic-circuit design, as Example 3-18 illustrates.

EXAMPLE 3-18

In a certain manufacturing process, a conveyor belt will shut down whenever specific conditions occur. These conditions are monitored and reflected